Quantification: 535_z27c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	19
interior lacuna count	17
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 17)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3454.53	2231	2978.24	px^2
major_axis_length_px	lacuna shape	93.8537	94.2448	42.5272	px
minor_axis_length_px	lacuna shape	48.5015	45.2188	21.6527	px
aspect_ratio	lacuna shape	2.0031	2.1668	0.6893	unitless
eccentricity	lacuna shape	0.8115	0.8871	0.1338	unitless
solidity	lacuna shape	0.8015	0.7988	0.0992	unitless
roots_count	headline	7.7059	7	4.3123	count
ring_length_r30_px	headline	249.647	208	123.877	px
ring_length_r60_px	reported	544.941	457	284.214	px
owned_length_px	ownership-dependent	689.4	331.35	639.428	px
edge_count	ownership-dependent	30.3529	13	28.1335	count
mean_edge_length_px	ownership-dependent	23.8629	25.0696	6.8503	px
perimeter_px	normalised	305.888	254.072	156.512	px
ring_area_r30_px2	normalised	9719.47	9403	4293.3	px^2
ring_area_r60_px2	normalised	20650.6	19497	8911.77	px^2
in_frame_fraction_r30	normalised	0.9922	1	0.0321	unitless
in_frame_fraction_r60	normalised	0.985	1	0.0578	unitless
ring_density_r30	normalised	0.0255787	0.0248781	0.00637339	px^-1
ring_density_r60	normalised	0.0260453	0.0255106	0.00654398	px^-1
roots_per_100px_perimeter	normalised	2.6051	2.325	1.0192	per 100 px
ring_attached_length_r30_px	network v2	211.353	169	126.41	px
ring_attached_length_r60_px	network v2	374.235	254	256.766	px
ring_length_w_r30_px	network v2	282.542	230.095	140.044	px
ring_length_w_r60_px	network v2	616.568	522.002	325.786	px
sholl_crossings_r10	network v2	6.7059	5	4.1043	count
sholl_crossings_r20	network v2	7.4706	5	4.4738	count
sholl_crossings_r30	network v2	7.7059	6	4.5794	count
loop_count	loops, report only	1.4118	0	2.2929	count
loop_edge_share	loops, report only	0.0788	0	0.1208	share
ring_width_mean_r30_px	width	7.66	7.8006	1.2177	px

Per field
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measure	value	units
analysed_area_px2	987481	px^2
total_skeleton_length_px	24512	px
canalicular_length_density_per_px	0.0248228	px^-1
junction_count	430	count
junction_density_per_px2	0.00043545	px^-2
lacuna_count	19	count

lacunae_per_px2	1.924e-05	px^-2
skeleton_component_count	414	count
mean_component_length_px	59.2077	px
median_component_length_px	23	px
field_length_density_w_per_px	0.0279795	px^-1
field_density_without_flagged_per_px	0.0231623	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	1242	px
unattached_component_count	75	count
canalicular_length_density_attached_per_px	0.0248228	px^-1
field_length_density_with_unattached_per_px	0.0260805	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	31	count
loop_edge_share	0.0917	share
edge_count	1254	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.5266	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	19040	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2942	101.888	45.2188	2.2532	0.8961	0.7542	-1.1126	105.425	685.744	False
2	957	54.4409	24.7864	2.1964	0.8903	0.7988	0.0308	119.021	790.769	False
3	1484	55.1992	46.0237	1.1994	0.5521	0.6199	-0.2191	163.815	808.798	False
4	3599	102.364	47.1738	2.1699	0.8875	0.8955	-1.0474	233.848	107.369	False
5	1807	63.8235	38.7415	1.6474	0.7947	0.9372	1.5189	218.143	826.646	False
6	712	35.4872	28.8379	1.2306	0.5828	0.9368	-0.3495	301.131	1012.33	True
7	11051	179.07	107.127	1.6716	0.8013	0.7079	0.7029	439.405	253.828	False
8	5312	120.404	59.815	2.0129	0.8679	0.8459	-1.268	460.68	557.419	False
9	1257	48.9366	34.4065	1.4223	0.7111	0.929	1.2464	562.555	134.776	False
10	796	41.9622	30.4426	1.3784	0.6882	0.761	0.6839	561.568	949.678	False
11	2123	56.2494	49.4219	1.1381	0.4775	0.9222	0.3101	574.703	493.107	False
12	2990	94.2448	43.1713	2.183	0.8889	0.8771	0.9473	600.532	860.511	False
13	1556	70.3776	32.2729	2.1807	0.8887	0.7988	1.1709	626.462	984.626	False
14	1310	65.9258	27.032	2.4388	0.9121	0.8501	-1.1711	624.014	302.034	False
15	9402	173.447	80.0495	2.1668	0.8871	0.7022	-0.8455	691.761	675.898	False
16	2231	116.941	28.0885	4.1633	0.9707	0.7914	0.9981	758.602	307.963	False
17	6401	142.513	65.0238	2.1917	0.8898	0.8153	-1.2115	832.616	623.668	False
18	3509	107.726	65.7294	1.6389	0.7923	0.6187	-0.7908	851.443	291.978	False
19	1656	75.1349	31.8764	2.3571	0.9055	0.792	-0.7612	861.306	996.914	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	408	877	1151.55	69	16.6891	8.4707
2	4	121	224	134.439	5	26.8877	8.3237
3	7	208	337	255.924	11	23.2658	8.0191
4	6	284	834	1790.48	84	21.3152	7.1459
5	4	140	363	101.154	4	25.2886	7.8006
6	2	92	276	116.841	7	16.6915	7.7797
7	15	445	1043	1159.97	51	22.7446	7.7299
8	18	336	754	935.754	48	19.4949	8.3125
9	7	161	457	525.002	17	30.8825	8.5498
10	3	84	144	37.2132	5	7.4426	3.9876
11	5	180	387	295.794	9	32.866	8.4103
12	10	245	465	215.551	12	17.9626	6.5239
13	6	148	290	300.836	12	25.0696	9.9005
14	3	142	269	147.154	4	36.7886	7.8455
15	13	492	1008	2077.65	78	26.6366	7.5315
16	4	190	407	331.35	13	25.4885	7.2625
17	8	275	661	761.642	41	18.5766	7.2476
18	10	385	744	1498.33	53	28.2704	7.1584
19	5	125	247	272.066	9	30.2296	7.6324

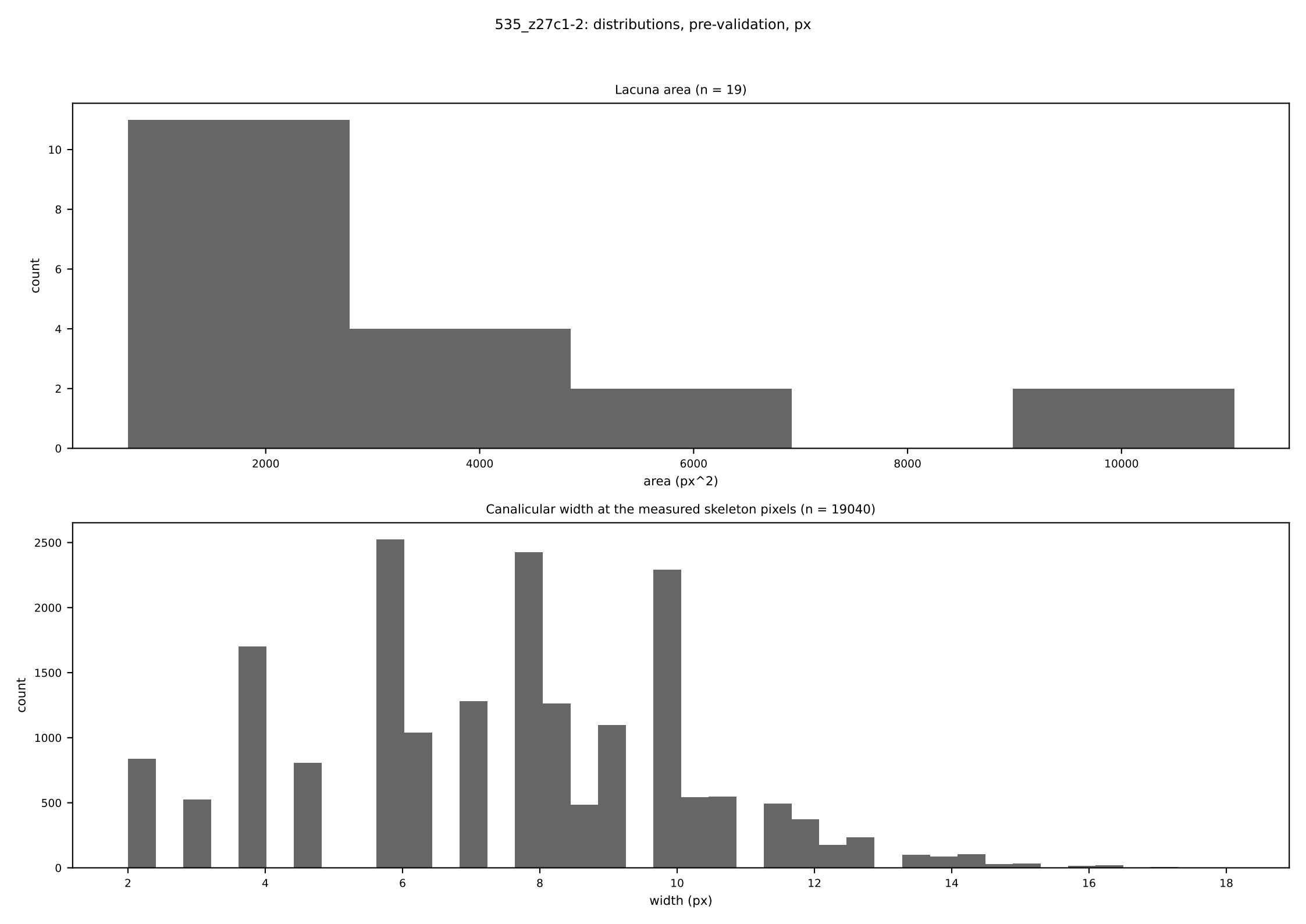
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	369.203	10657	23260	1	1	0.0382847	0.0377042	2.1668
2	159.095	4461	7968	1	1	0.027124	0.0281125	2.5142
3	254.072	5329	9525	1	1	0.0390317	0.0353806	2.7551
4	281.948	10538	26647	1	1	0.0269501	0.0312981	2.128
5	184.853	6087	14824	1	1	0.0229998	0.0244873	2.1639
6	112.426	3564	9950	0.5858	0.5585	0.0258137	0.0277387	1.7789
7	645.151	19717	40885	1	1	0.0225694	0.0255106	2.325
8	368.818	12630	27408	1	1	0.0266033	0.0275102	4.8805
9	147.054	6972	18793	1	1	0.0230924	0.0243176	4.7602
10	153.338	5451	11094	1	0.9843	0.01541	0.01298	1.9565
11	184.267	7991	18400	1	1	0.0225253	0.0210326	2.7135
12	250.392	9403	19497	1	1	0.0260555	0.0238498	3.9937
13	219.35	5949	9604	0.8676	0.7611	0.0248781	0.0301958	2.7354
14	195.48	7849	18629	1	1	0.0180915	0.0144399	1.5347
15	634.625	17491	33867	1	1	0.0281288	0.0297635	2.0485
16	284.534	9498	19681	1	1	0.0200042	0.0206798	1.4058
17	465.843	13368	26997	1	1	0.0205715	0.0244842	1.7173
18	402.073	11840	23982	1	1	0.0325169	0.0310233	2.4871
19	212.551	6244	14852	0.731	0.6597	0.0200192	0.0166308	2.3524

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	382	678	461.818	987.607	10	16	14
2	110	110	140.125	254.137	5	4	2
3	162	228	230.095	376.978	4	7	5
4	224	543	320.108	931.725	6	8	11
5	90	99	161.853	406.576	3	4	6
6	85	107	98.9411	303.622	3	3	3
7	381	718	507.902	1193.57	14	15	15
8	298	631	369.907	837.683	9	10	9
9	157	331	186.823	522.002	5	5	6

10	24	24	100.296	163.752	1	2	1
11	139	231	199.61	432.433	4	5	8
12	174	232	281.007	525.245	6	7	5
13	97	162	169.61	327.25	3	5	3
14	107	184	157.297	299.279	4	5	4
15	482	888	560.659	1174.86	15	16	15
16	169	254	210.539	456.848	5	5	6
17	255	444	310.693	748.884	13	3	7
18	342	605	434.877	842.826	7	10	14
19	107	198	136.983	277.622	6	4	3


535_z27c1-2: distributions, pre-validation, px
Lacuna area (n = 19)
count
area (px^2)
0
2
4
6
8
10
2000
4000
6000
8000
10000
Canalicular width at the measured skeleton pixels (n = 19040)
count
width (px)
0
500
1000
1500
2000
2500
2
4
6
8
10
12
14
16
18